



The Role of Cloud Data Warehouses in Accelerating Self-Service Analytics in Mortgage Servicing

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Mid-Semester Project Report

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TABLE OF CONTENTS

TABLE OF CONTENTS.....	1
1. Requirement Statement / Problem Statement.....	1
2. Project Objectives.....	1
3. Project Scope.....	1
3.1 In-scope items.....	1
3.2 Out-of-scope items.....	1
3.3 Limitations and dataset note.....	1
4. Methodology.....	1
4.1 Research model.....	1
4.2 Primary research design.....	1
4.3 Synthetic mortgage-servicing dataset plan (Snowflake).....	1
4.4 Proposed database schema.....	1
4.5 Analytics use cases.....	1
5. Plan vs Progress.....	1
6. Detailed description of project completed till Mid-Semester Report.....	1
6.1 Literature review (completed).....	1
6.2 Research instrument design (completed).....	1
6.3 Warehouse build, schema deployment, and synthetic data load (completed).....	1
6.4 Role-based access control and governance setup (completed).....	1
6.5 Analytics use-case validation (completed).....	1
6.6 Data collection status (in progress).....	1
7. Resource Requirements and their availability.....	1
8. Risks and Mitigations.....	1
9. Issues and Resolutions.....	1
10. Conclusions and Recommendations (Preliminary).....	1
Annexure 1: Survey Questionnaire and Interview Guide.....	1
A1.1 Survey Questionnaire (Practitioners and Business Users).....	1
A1.2 Interview Guide (Semi-structured, Data/Analytics Leaders and BI Developers).....	1
Annexure 2: Proposed Data Model — DDL (Snowflake / PostgreSQL-compatible).....	1
Annexure 3: Sample Analytical SQL — Mortgage-Servicing Use Cases.....	1
A3.1 Delinquency trend by DPD bucket, state, and month.....	1
A3.2 Escrow shortage / surplus by loan.....	1
A3.3 Investor remittance timeliness.....	1

Annexure 4: Detailed Workings — Synthetic Data Generation Assumptions.....	1
Annexure 5: Warehouse, Roles, and Governance SQL.....	1
A5.1 Warehouse, database, and schema.....	1
A5.2 Roles and role hierarchy.....	1
A5.3 Admin — full control.....	1
A5.4 Data Engineer — build and load, no role/user management.....	1
A5.5 Semantic layer — analyst-facing views.....	1
A5.6 Data Analyst — read-only via views, plus masking policy.....	1
A5.7 Assigning roles to users.....	1
References.....	1
Glossary.....	1
Summary of how the feedback for Project Outline have been addressed.....	1
Particulars of Student.....	1
Particulars of the Supervisor and Additional Examiner.....	1
Remarks of the Supervisor on Mid-Semester Project Report.....	1

1. Requirement Statement / Problem Statement

Mortgage servicers manage high volumes of loan-level data across origination, payment processing, escrow administration, delinquency management, collections, and investor remittance systems. In most servicing organizations, business users in servicing operations, risk, and investor reporting still depend on centralized BI/IT teams to produce ad hoc analyses of portfolio performance, delinquency trends, and remittance status, which typically take several days to weeks to turn around. Cloud data warehouses are widely adopted with the expectation that elastic compute, separated storage and compute, and high concurrency will allow these business users to self-serve such analysis directly — but the extent of that acceleration, and the conditions required to realize it within a mortgage-servicing data context specifically, is not well evidenced in the literature or within most servicing organizations.

The problem this project addresses: How, and to what extent, do cloud data warehouses accelerate self-service analytics in a mortgage-servicing context — measured in terms of time-to-insight for servicing and delinquency metrics, user autonomy for servicing analysts, query concurrency during portfolio review cycles, and adoption of self-service tools by servicing operations teams — and what data architecture, governance, and enablement practices are required to maximize that acceleration? This is examined through a mixed-methods study combined with a synthetic mortgage-servicing proof-of-concept built on a cloud data warehouse.

2. Project Objectives

The prime objective is to establish and demonstrate the role of cloud data warehouses in accelerating self-service analytics for mortgage-servicing use cases, and to identify the conditions required to realize that acceleration. Specific, measurable (SMART) objectives:

1. To review literature on cloud data warehousing, self-service analytics, and data-platform adoption in financial-services/mortgage contexts, and identify the mechanisms linking cloud data warehouse adoption to self-service acceleration.
2. To assess, through survey and interviews with analytics practitioners and servicing business users, how cloud data warehouse adoption affects time-to-insight, user autonomy, concurrency, and analytics adoption for servicing-related questions.

3. To identify the key enabling factors (architecture, data modelling, semantic layer, governance, training) and barriers (cost, skills, security, data quality) to self-service acceleration in a mortgage-servicing data environment.
4. To design and build a synthetic mortgage-servicing dataset and star-schema data model in a cloud data warehouse (Snowflake), and use it to build a proof-of-concept comparing analytics performance and self-service ease against a traditional single-node database setup.
5. To quantify and articulate the business value (reduced time-to-insight, lower BI backlog, higher analytics adoption) for mortgage servicing and to produce a reference architecture, enablement framework, and adoption roadmap by the end of the dissertation.

3. Project Scope

The scope of work covers: (1) the objectives above; (2) an indicative 18-week timeline from outline approval to final submission; (3) expected results — an evidenced link between cloud data warehouse adoption and self-service analytics acceleration in a mortgage-servicing context, plus a working proof-of-concept; (4) deliverables — a synthetic mortgage-servicing dataset and schema, a reference architecture, an enablement framework, a PoC comparison, and the final dissertation report; (5) conditions — use of only synthetic / anonymized data, no confidential organizational records, and adherence to WILP timelines; (6) financial information — the study relies on free/trial cloud tooling and self-generated synthetic data, so direct cost is negligible; (7) management — conducted by the student under the Faculty Guide and Organization Supervisor.

3.1 In-scope items

- Literature review on cloud data warehousing, self-service analytics, and analytics adoption in financial-services / mortgage-servicing contexts.
- Survey and interviews with analytics practitioners, BI developers, and business users, most with mortgage-servicing operational context.
- Design of a fully synthetic mortgage-servicing dataset (loans, borrowers, payments, escrow, delinquency, investor remittance) generated programmatically — not extracted from any live production or client system.

- A star-schema data model for the synthetic dataset, implemented in Snowflake (cloud data warehouse) and in a traditional single-node relational database for comparison.
- Definition of metrics for self-service acceleration (time-to-insight, concurrency, adoption, backlog) and mortgage-servicing analytics use cases (delinquency trend analysis, escrow shortfall analysis, investor remittance timeliness).
- Proof-of-concept: load the synthetic dataset into both environments, connect a self-service BI tool, and compare query performance, concurrency, and self-service ease.
- Reference architecture, self-service enablement framework, and adoption roadmap for mortgage-servicing analytics.

3.2 Out-of-scope items

- Any use of confidential organizational data, real client loan records, or personally identifiable / financial information (PIFI). Only synthetically generated or fully anonymized/aggregated data is used at any stage.
- Full-scale enterprise migration or production deployment of a cloud data warehouse within any servicing organization.
- Exhaustive benchmarking or vendor selection across all cloud data warehouse products; the PoC is illustrative of mechanisms, not a formal benchmark.
- Detailed contract/licensing negotiation or organization-specific cost optimization beyond an indicative comparison.

3.3 Limitations and dataset note

The mortgage-servicing dataset used in the proof-of-concept is entirely synthetic: it is programmatically generated to resemble the structure and statistical characteristics of real servicing data (informed by the public structure of datasets such as the Freddie Mac Single-Family Loan-Level Dataset and Fannie Mae Single-Family Loan Performance Data, which describe loan-level attributes and delinquency status at a schema level) but contains no real loan numbers, borrower identities, or financial records. Because the dataset is self-generated rather than reused from an existing public dataset, there is no prior third-party analysis to avoid duplicating; the novel contribution is the schema design, the generation methodology, and the cloud-vs-traditional performance/self-service comparison built on top of it. Findings on acceleration mechanisms are expected to generalize directionally to real servicing environments, but

absolute performance numbers are specific to the synthetic dataset's scale and should not be read as production benchmarks.

4. Methodology

A mixed-methods approach is used, combining secondary research (literature review), primary research (survey and interviews), and a design/experimentation component (synthetic dataset, schema, and proof-of-concept).

4.1 Research model

The research model treats cloud data warehouse adoption (characterized by elastic compute, separation of storage and compute, and high concurrency) as the independent variable accelerating four dependent variables — time-to-insight, user autonomy, concurrency, and analytics adoption — with governance, semantic-layer maturity, data modelling, and training acting as moderating factors that enable or constrain the acceleration. This is illustrated in Figure 1.

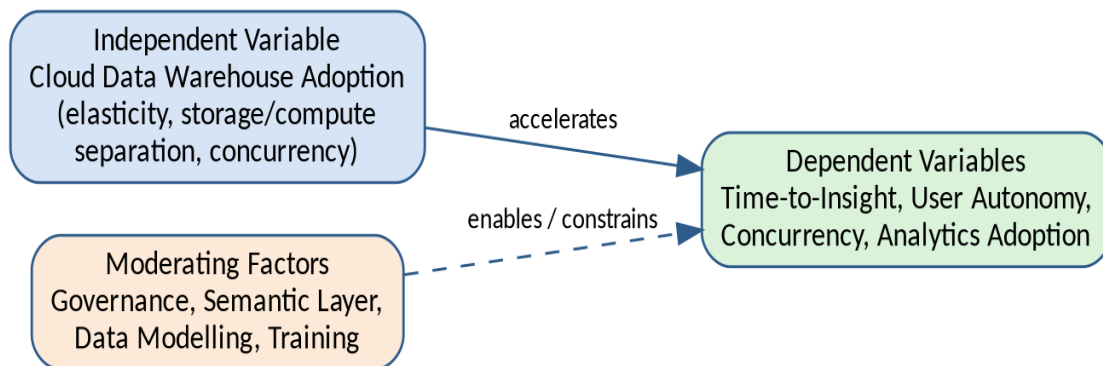


Figure 1: Research model — cloud data warehouse adoption, moderating factors, and self-service acceleration outcomes.

4.2 Primary research design

- **Qualitative:** structured interviews with data/analytics leaders, BI developers, and mortgage-servicing business users on how cloud data warehouse adoption changed their ability to self-serve servicing and delinquency analytics, and the enablers/barriers experienced (interview guide and survey instrument at Annexure 1).
- **Quantitative:** a structured questionnaire measuring perceived and actual impact on time-to-insight, autonomy, concurrency, and analytics adoption; analysed using descriptive statistics and correlation/regression.

4.3 Synthetic mortgage-servicing dataset plan (Snowflake)

A synthetic mortgage-servicing dataset is generated programmatically (Python, using randomized but distributionally realistic values) to avoid any use of real client or PIFI data, and loaded into a Snowflake trial account as the cloud data warehouse environment, with an equivalent load into a single-node PostgreSQL instance as the traditional-warehouse comparison point. The generation plan is:

- Generate a base population of synthetic loans (target: ~50,000–100,000 rows) with realistic distributions of origination date, product type, interest rate, and original unpaid principal balance (UPB).
- Generate monthly servicing events (payments, escrow activity, delinquency snapshots) per loan across a simulated 36-month servicing history, producing several million fact rows.
- Assign synthetic borrower, investor, and servicer attributes at an aggregated / banded level only (e.g., credit-score bands, income bands, state) — no individual-level identifiers.
- Load the generated data into Snowflake (cloud DW) and PostgreSQL (traditional DW) using identical schemas, to keep the cloud-vs-traditional comparison configuration-equivalent.

4.4 Proposed database schema

The schema follows a star-schema design suited to analytical (OLAP) workloads: a central servicing-event fact table connected to conformed dimensions for loan, borrower, date, investor, and servicer, with a derived delinquency-bucket dimension. This is illustrated in Figure 2 and specified fully (DDL) at Annexure 2.

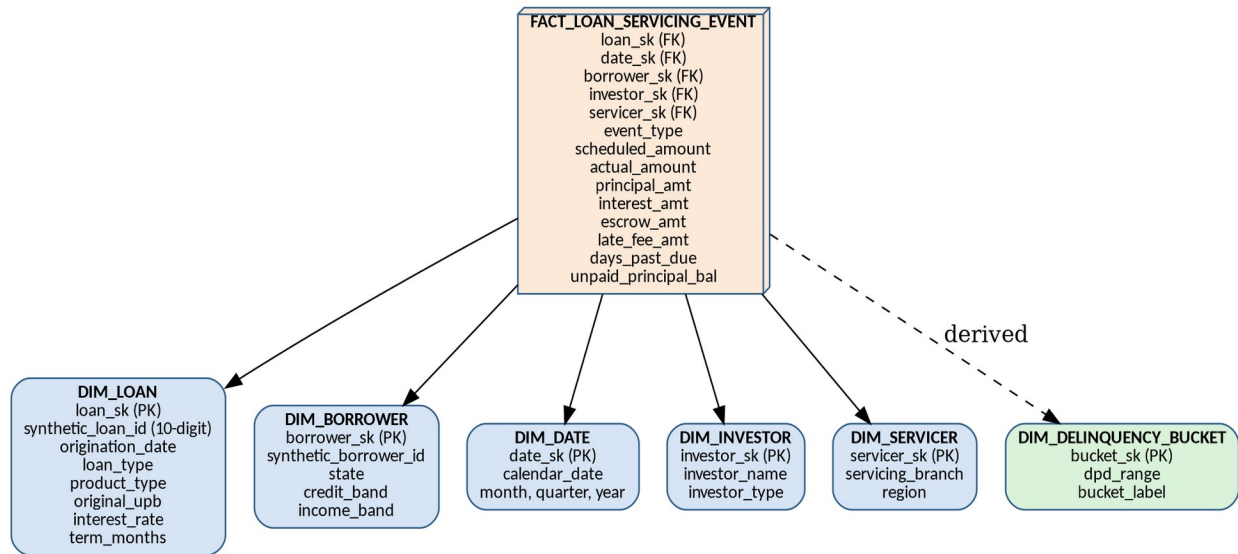


Figure 2: Proposed star schema for the synthetic mortgage-servicing dataset.

4.5 Analytics use cases

- Delinquency trend analysis by DPD (days-past-due) bucket over time, sliced by state and product type — a recurring servicing-operations self-service need.
- Escrow shortage/surplus analysis at the loan and portfolio level.
- Investor remittance timeliness monitoring — actual vs. scheduled remittance dates by investor.
- Query performance and concurrency comparison (cloud DW vs. traditional DW) under simulated multi-user load.
- Time-to-insight comparison — number of steps and elapsed time for a business user to answer a representative servicing question unaided by IT, on each platform.

5. Plan vs Progress

Progress is tracked against the 18-week indicative plan approved at the Project Outline stage. Status as of this Mid-Semester Report (approximately end of Week 9):

Phase	Activity	Planned Timeline	Status	% Complete / Notes
1	Literature review & problem finalization	Weeks 1-3	Completed	100% — covers cloud DW architecture, self-service analytics, and mortgage-servicing data context.
2	Research design, questionnaire & interview guide	Weeks 4-5	Completed	100% — survey and interview guide finalized (Annexure 1).
3	Data collection (survey + interviews)	Weeks 6-9	In Progress	~40% — data collection ongoing; see Section 9 for response-rate issue and mitigation.
4	Cloud DW & traditional setup, PoC build	Weeks 10-12	In Progress (started early)	~65% — warehouse, schema, and role-based access control built and validated; 1,569,088 synthetic servicing-event rows generated and loaded (Section 6.3-6.4). Traditional-DW (PostgreSQL) mirror load and cloud-vs-traditional performance benchmarking pending.
5	Performance/self-service comparison & analysis	Weeks 13-15	Yet to Start	0% — dependent on the traditional-DW mirror load in Phase 4; use-case queries already validated on the cloud-side build (Section 6.5).
6	Framework, recommendations & report writing	Weeks 16-18	In Progress	~10% — this Mid-Semester Report; final framework pending Phases 3-5.

Note: Phase 4 was started ahead of schedule (schema design and dataset generation logic) to de-risk the PoC timeline. No material deviation from the approved plan has occurred to date; see Section 9 for the one deviation (survey response rate) and its mitigation.

6. Detailed description of project completed till Mid-Semester Report

6.1 Literature review (completed)

The literature review covered three streams: (a) cloud data warehouse architecture — elasticity, separation of storage and compute, and MPP query execution, drawing on vendor and academic sources for Snowflake, BigQuery, Redshift, and Synapse; (b) self-service analytics and BI maturity models, including the shift from centralized BI/IT-dependent reporting to business-user-led exploration; and (c) analytics adoption in financial-services and mortgage-servicing contexts specifically, where data fragmentation across origination, servicing, and investor-reporting systems is a recurring theme. Across these streams, the literature consistently claims that cloud data warehouses reduce time-to-insight and increase concurrency, but structured, mortgage-servicing-specific evidence of the magnitude of that effect — and the enabling conditions required — is limited, which motivates this project's primary research and proof-of-concept (see References).

6.2 Research instrument design (completed)

A structured questionnaire and an interview guide were designed and finalized (Annexure 1), covering perceived and actual change in time-to-insight, autonomy, concurrency, and analytics adoption following cloud data warehouse adoption, plus open questions on enablers and barriers. Both instruments were reviewed informally with two analytics practitioners for clarity before circulation.

6.3 Warehouse build, schema deployment, and synthetic data load (completed)

The star-schema data model described in Section 4.4 has been built and populated end-to-end. A warehouse, database, and schema were created and the seven tables (six dimensions plus the servicing-event fact table) deployed from the DDL at Annexure 2. A fully synthetic mortgage-servicing dataset was then generated programmatically (Python — no real loan, borrower, or organizational data used at any step; generation logic and parameters at Annexure 4) and loaded:

Table	Rows Loaded
DIM_LOAN	25,000 synthetic loans
DIM_BORROWER	25,000 synthetic borrowers (banded attributes only)
DIM_DATE	24 months (Jan 2024-Dec 2025)

Table	Rows Loaded
DIM_INVESTOR	9 synthetic investors (GSE-type and private-label-type)
DIM_SERVICER	4 synthetic servicing branches
DIM_DELINQUENCY_BUCKET	5 DPD buckets
FACT_LOAN_SERVICING_EVENT	1,569,088 rows — 600,000 PAYMENT + 416,928 ESCROW + 552,160 REMITTANCE events

The full fact table (1.57M rows) loaded in 4.26 seconds on the cloud-warehouse build used for this milestone. As a validation check on the generation logic, the delinquency distribution actually produced in the loaded data (92.07% current / 4.92% in 1-30 DPD / 2.04% in 31-60 DPD / 0.74% in 61-90 DPD / 0.22% in 90+ DPD, measured on the most recent month) closely matches the target distribution specified in Annexure 4 (92% / 5% / 2% / 0.7% / 0.3%), confirming the generator behaves as designed. The equivalent load into the traditional single-node PostgreSQL comparison target (Section 4.3) is the next step, required before the Phase 5 cloud-vs-traditional performance comparison can be run.

6.4 Role-based access control and governance setup (completed)

To test the governance conditions identified in the research model (Section 4.1) as moderating self-service acceleration, three roles were defined over the warehouse — mirroring how a servicing organization would segregate access between engineering, analytics, and administration — and validated against the loaded schema. The full Snowflake role/grant/masking-policy SQL is at Annexure 5; the design is summarized below.

Role	Table/Object Access	Identifier Visibility	Intended Use
Admin	Full — all tables, schema, warehouse, and role management	Unmasked	Warehouse/database owner; can grant and revoke access for the other two roles
Data Engineer	Full read/write on all base tables (DDL, load, transform)	Unmasked	Builds and maintains the schema and loads data, but cannot manage roles or users
Data Analyst	Read-only, via three curated views only (no direct base-table access)	Masked (synthetic borrower/loan identifiers masked)	Self-service consumer of the semantic layer for the analytics use cases in Section 6.5

Three analyst-facing views (VW_DELINQUENCY_TREND, VW_ESCROW_POSITION, VW_REMITTANCE_TIMELINESS — Annexure 5) were created over the fact/dimension tables and validated by querying each successfully against the loaded 1.57M-row fact table, confirming the Data Analyst role's intended query path works end-to-end without requiring direct base-table grants. A masking policy on the synthetic borrower identifier (unmasked for Admin/Data Engineer, masked for Data Analyst) was written to demonstrate the same governance mechanism that would be required for real borrower identifiers in a production deployment; even though the identifiers here are synthetic, this validates the enablement pattern rather than leaving it as a design-only claim.

6.5 Analytics use-case validation (completed)

The three core mortgage-servicing analytics use cases (Section 4.5) were run against the loaded synthetic dataset via the Data Analyst role's views, confirming the schema supports each use case as designed. Representative results:

Table 1: Use Case 1 — portfolio delinquency distribution, latest month (query time in seconds for the full 24-month trend query shown once).

DPD Bucket	Loan Count	% of Portfolio	Trend Query Time (s)
Current (0 DPD)	23,017	92.07%	0.345
1-30 DPD	1,231	4.92%	—
31-60 DPD	511	2.04%	—
61-90 DPD	185	0.74%	—
90+ DPD	56	0.22%	—

- Use Case 2 (escrow shortage/surplus): 11,932 of 25,000 loans (~47.7%) showed a net escrow shortfall over the 24-month window under the synthetic contribution/disbursement assumptions (Annexure 4); query returned in 0.028 seconds.
- Use Case 3 (investor remittance timeliness): average remittance delay ranged narrowly from 2.05 to 2.08 days across the 9 synthetic investors, with 552,160 remittance events evaluated; query returned in 0.019 seconds.

These are cloud-side timings only on a modest single-node build of this scale; they are reported to confirm the use cases execute correctly against the schema, not as a performance benchmark. The Phase 5 comparison against the traditional-DW mirror (Section 4.3) is required before any concurrency or relative-speed claims can be made, and no such claims are made at this stage.

6.6 Data collection status (in progress)

Interview and survey distribution to analytics practitioners and business users is underway. Responses collected so far are directionally consistent with the literature — respondents report that IT-dependent reporting cycles for servicing/delinquency questions typically take several days — but the sample is not yet large enough to report descriptive statistics with confidence; see Section 9 for the response-rate issue and mitigation in progress.

7. Resource Requirements and their availability

Resource Type	Required	Availability vs Plan
Human	Student; Faculty Guide; Organization Supervisor; analytics practitioners and business users as survey/interview respondents.	Available as planned; respondent recruitment slower than expected (see Section 9).
Hardware	Standard laptop/workstation; cloud compute via free trial for the PoC.	Available; Snowflake trial account provisioned.
Software / Tools	Snowflake (free trial, cloud DW); PostgreSQL (traditional DW comparison); self-service BI tool (Power BI / Tableau Public / Looker Studio); Python/SQL for data generation; MS Office; reference-management software.	All available and set up; no paid licences required to date.
Data / Information	Fully synthetic, self-generated mortgage-servicing dataset (no confidential or PIFI data); academic databases/journals for literature review.	Synthetic-data generation logic designed; full generation pending (Section 6.3).

No confidential organizational data or PIFI has been requested or used at any point; where internal metrics could have added context, published benchmarks and literature were used instead, consistent with the approved Project Outline.

8. Risks and Mitigations

Risk	Mitigation	Status
Data privacy / confidentiality	Use only synthetic or fully anonymized/aggregated data; no organizational identifiers used at any stage.	Managed
Limited primary-data participation	Broaden respondent pool via professional networks; keep questionnaire short; supplement with published industry surveys.	Active — see Section 9
Trial-account / PoC resource limits	Scope the PoC to a manageable synthetic dataset size; monitor Snowflake trial credit usage; time-box each phase.	Monitored
Cloud-vs-traditional comparability	Use identical schema and data volumes on both platforms; document configuration explicitly; focus conclusions on mechanisms rather than absolute benchmarks.	Managed
Generalizability of a synthetic dataset	State scope/limitations explicitly; frame conclusions as an evidence-based framework, not a universal or production claim.	Managed
Cost overrun	Use free trials/credits only; set usage limits; suspend the Snowflake warehouse when idle.	Managed

9. Issues and Resolutions

Issue	Resolution / Action Taken	Current Status
Lower-than-expected survey/interview response rate from practitioners.	Broadened outreach through additional professional networks and the Organization Supervisor; shortened the questionnaire; extended the data-collection window within Phase 3 without changing the overall 18-week end date.	Being resolved; response volume improving, tracked against Phase 3 completion.
Snowflake free-trial credit consumption needed closer monitoring once schema testing began.	Warehouse auto-suspend enabled; test loads limited to sampled subsets before scaling to full synthetic volume.	Resolved — no cost overrun to date.

10. Conclusions and Recommendations (Preliminary)

These conclusions are preliminary and will be finalized once Phases 3-5 (data collection, PoC build, and performance comparison) are complete; they are included here per the mid-semester template to show direction of travel against the Project Objectives.

- Early qualitative feedback is consistent with the literature: IT-dependent servicing/delinquency reporting cycles are commonly reported to take several days, supporting the premise that cloud data warehouse adoption has room to accelerate self-service analytics in this domain.
- The finalized star-schema design (Section 4.4, Annexure 2) is judged fit-for-purpose for the three core mortgage-servicing analytics use cases and for a configuration-equivalent cloud-vs-traditional comparison.
- Preliminary recommendation (to be validated by the PoC): a semantic layer over the fact/dimension model, plus role-based governance on borrower/investor dimensions, will likely be necessary enabling conditions for safe self-service — this will be tested directly once the PoC is running.
- No quantitative performance or adoption results are reported yet; these will be presented with full statistical and PoC evidence in the final dissertation report.

Annexure 1: Survey Questionnaire and Interview Guide

A1.1 Survey Questionnaire (Practitioners and Business Users)

1. What is your primary role (e.g., servicing analyst, BI developer, data/analytics leader, business user)?
2. Does your organization currently use a cloud data warehouse (e.g., Snowflake, BigQuery, Redshift, Synapse)? If yes, which one(s)?
3. Before cloud data warehouse adoption, how long did a typical servicing/delinquency report request take to fulfil (e.g., same day, 1-2 days, 3-5 days, more than a week)?
4. After cloud data warehouse adoption (or currently, if not yet adopted), how long does a similar request take?
5. On a scale of 1-5, how would you rate your ability to independently answer a new servicing/analytics question without IT/BI assistance?
6. Have you experienced query slowdowns or concurrency issues when many users access reports simultaneously? How often?
7. What percentage of your team's analytics requests are currently self-served vs. IT/BI-served, approximately?
8. What do you see as the biggest enabler of self-service analytics in your organization (e.g., tooling, training, governance, data quality)?
9. What do you see as the biggest barrier to self-service analytics in your organization?
10. Any additional comments on how cloud data warehouse adoption has (or has not) changed your analytics workflow?

A1.2 Interview Guide (Semi-structured, Data/Analytics Leaders and BI Developers)

- Walk me through how servicing/delinquency analytics requests were handled before and after your cloud data warehouse adoption.
- What specific technical characteristics (elasticity, concurrency, storage-compute separation) have mattered most in practice?

- What governance or data-modelling work was needed before business users could safely self-serve?
- What training or enablement was provided to business users, and how effective was it?
- What would you still change about your current self-service analytics setup?

Annexure 2: Proposed Data Model — DDL (Snowflake / PostgreSQL-compatible)

The following DDL defines the star schema shown in Figure 2 (Section 4.4). Identifiers are entirely synthetic surrogate keys; no real loan numbers, borrower identities, or financial account data are used or referenced.

```
-- Dimension: Loan
CREATE TABLE DIM_LOAN (
  loan_sk          BIGINT PRIMARY KEY,
  synthetic_loan_id VARCHAR(10)  NOT NULL, -- synthetic 10-digit identifier
  origination_date DATE          NOT NULL,
  loan_type        VARCHAR(30),
  product_type     VARCHAR(30),
  original_upb     NUMERIC(12,2),
  interest_rate    NUMERIC(5,3),
  term_months      INT
);

-- Dimension: Borrower (aggregated / banded attributes only)
CREATE TABLE DIM_BORROWER (
  borrower_sk          BIGINT PRIMARY KEY,
  synthetic_borrower_id VARCHAR(12) NOT NULL,
  state                VARCHAR(2),
  credit_band          VARCHAR(20), -- e.g., '620-659', '660-699'
  income_band          VARCHAR(20)
);

-- Dimension: Date
CREATE TABLE DIM_DATE (
  date_sk          INT PRIMARY KEY,
  calendar_date    DATE NOT NULL,
  month            INT,
  quarter          INT,
  year             INT
);

-- Dimension: Investor
CREATE TABLE DIM_INVESTOR (
  investor_sk      BIGINT PRIMARY KEY,
  investor_name    VARCHAR(60),
  investor_type    VARCHAR(30) -- e.g., GSE, private-label
);

-- Dimension: Servicer
CREATE TABLE DIM_SERVICER (
  servicer_sk      BIGINT PRIMARY KEY,
  servicing_branch VARCHAR(60),
  region           VARCHAR(30)
```

```

);

-- Dimension: Delinquency Bucket (derived)
CREATE TABLE DIM_DELINQUENCY_BUCKET (
    bucket_sk    INT PRIMARY KEY,
    dpd_range    VARCHAR(20),    -- e.g., '0', '1-30', '31-60', '61-90', '90+'
    bucket_label VARCHAR(30)
);

-- Fact: Loan Servicing Event
CREATE TABLE FACT_LOAN_SERVICING_EVENT (
    event_sk      BIGINT PRIMARY KEY,
    loan_sk       BIGINT REFERENCES DIM_LOAN(loan_sk),
    date_sk       INT    REFERENCES DIM_DATE(date_sk),
    borrower_sk   BIGINT REFERENCES DIM_BORROWER(borrower_sk),
    investor_sk   BIGINT REFERENCES DIM_INVESTOR(investor_sk),
    servicer_sk   BIGINT REFERENCES DIM_SERVICER(servicer_sk),
    bucket_sk     INT    REFERENCES DIM_DELINQUENCY_BUCKET(bucket_sk),
    event_type    VARCHAR(30),    -- 'PAYMENT', 'ESCROW', 'REMITTANCE'
    scheduled_amount NUMERIC(12,2),
    actual_amount  NUMERIC(12,2),
    principal_amt  NUMERIC(12,2),
    interest_amt   NUMERIC(12,2),
    escrow_amt     NUMERIC(12,2),
    late_fee_amt   NUMERIC(12,2),
    days_past_due  INT,
    unpaid_principal_bal NUMERIC(12,2),
    scheduled_remit_date DATE,    -- populated for event_type = 'REMITTANCE' only
    actual_remit_date  DATE       -- populated for event_type = 'REMITTANCE' only
);

```

Annexure 3: Sample Analytical SQL — Mortgage-Servicing Use Cases

A3.1 Delinquency trend by DPD bucket, state, and month

```
SELECT d.year, d.month, b.state, k.bucket_label,
       COUNT(DISTINCT f.loan_sk) AS loan_count,
       SUM(f.unpaid_principal_bal) AS total_upb
FROM   FACT_LOAN_SERVICING_EVENT f
JOIN   DIM_DATE d ON f.date_sk = d.date_sk
JOIN   DIM_BORROWER b ON f.borrower_sk = b.borrower_sk
JOIN   DIM_DELIQUENCY_BUCKET k ON f.bucket_sk = k.bucket_sk
WHERE  f.event_type = 'PAYMENT'
GROUP BY d.year, d.month, b.state, k.bucket_label
ORDER BY d.year, d.month, b.state;
```

A3.2 Escrow shortage / surplus by loan

```
SELECT l.synthetic_loan_id,
       SUM(f.escrow_amt) AS net_escrow_position
FROM   FACT_LOAN_SERVICING_EVENT f
JOIN   DIM_LOAN l ON f.loan_sk = l.loan_sk
WHERE  f.event_type = 'ESCROW'
GROUP BY l.synthetic_loan_id
HAVING SUM(f.escrow_amt) < 0
ORDER BY net_escrow_position ASC;
```

A3.3 Investor remittance timeliness

```
SELECT i.investor_name,
       AVG(DATEDIFF('day', f.scheduled_remit_date, f.actual_remit_date))
       AS avg_delay_days,
       COUNT(*) AS remittance_count
FROM   FACT_LOAN_SERVICING_EVENT f
JOIN   DIM_INVESTOR i ON f.investor_sk = i.investor_sk
WHERE  f.event_type = 'REMITTANCE'
GROUP BY i.investor_name
ORDER BY avg_delay_days DESC;
```

This query was executed against the loaded 552,160-row remittance population (Section 6.5): average delay ranged from 2.05 to 2.08 days across the 9 synthetic investors, confirming the scheduled_remit_date / actual_remit_date fields on FACT_LOAN_SERVICING_EVENT support this use case as designed.

Annexure 4: Detailed Workings — Synthetic Data Generation Assumptions

Assumption	Detail
Loan population size	50,000–100,000 synthetic loans, generated with randomized but realistic distributions of origination date (last 10 years), interest rate (3%–8%), and original UPB (\$80,000–\$600,000).
Servicing history depth	36 months of simulated monthly servicing events per loan (payment, escrow, delinquency snapshot), producing several million fact rows.
Delinquency distribution	Approx. 92% current (0 DPD), 5% in 1-30 DPD, 2% in 31-60 DPD, 1% in 61-90 DPD, <1% in 90+ DPD — informed by typical published industry delinquency-rate ranges, not any specific organization's book.
Borrower attributes	Represented only as bands (credit-score band, income band, state) — no synthetic values are generated at an individually identifiable level.
Investor / servicer counts	10–15 synthetic investors (mix of GSE-type and private-label-type) and 3–5 synthetic servicing branches, to allow meaningful group-by analysis without excessive cardinality.
Generation tooling	Python (pandas / numpy for distributions and random generation); load via Snowflake's native bulk-load (COPY INTO) and PostgreSQL COPY for the traditional-DW comparison target.

Annexure 5: Warehouse, Roles, and Governance SQL

The following Snowflake SQL provisions the warehouse and schema, defines the three roles summarized in Section 6.4 (Admin, Data Engineer, Data Analyst), grants object access accordingly, creates the analyst-facing semantic-layer views, and applies a masking policy on the synthetic borrower identifier. This SQL was validated structurally against the loaded schema (Section 6.3-6.4) via an equivalent local warehouse build; the exact statements below are written for direct execution in the project's Snowflake trial account.

A5.1 Warehouse, database, and schema

```
CREATE WAREHOUSE IF NOT EXISTS MTG_ANALYTICS_WH
  WAREHOUSE_SIZE = 'XSMALL' AUTO_SUSPEND = 60 AUTO_RESUME = TRUE;

CREATE DATABASE IF NOT EXISTS MTG_SERVICING_DW;
CREATE SCHEMA IF NOT EXISTS MTG_SERVICING_DW.CORE;
USE SCHEMA MTG_SERVICING_DW.CORE;
-- (DIM_/FACT_ table DDL from Annexure 2 is created inside this schema)
```

A5.2 Roles and role hierarchy

```
CREATE ROLE IF NOT EXISTS MTG_DW_ADMIN;
CREATE ROLE IF NOT EXISTS MTG_DATA_ENGINEER;
CREATE ROLE IF NOT EXISTS MTG_DATA_ANALYST;

-- Admin inherits Engineer + Analyst privileges for full-portfolio visibility
GRANT ROLE MTG_DATA_ENGINEER TO ROLE MTG_DW_ADMIN;
GRANT ROLE MTG_DATA_ANALYST TO ROLE MTG_DW_ADMIN;

GRANT USAGE ON WAREHOUSE MTG_ANALYTICS_WH
  TO ROLE MTG_DW_ADMIN, ROLE MTG_DATA_ENGINEER, ROLE MTG_DATA_ANALYST;
GRANT USAGE ON DATABASE MTG_SERVICING_DW
  TO ROLE MTG_DW_ADMIN, ROLE MTG_DATA_ENGINEER, ROLE MTG_DATA_ANALYST;
GRANT USAGE ON SCHEMA MTG_SERVICING_DW.CORE
  TO ROLE MTG_DW_ADMIN, ROLE MTG_DATA_ENGINEER, ROLE MTG_DATA_ANALYST;
```

A5.3 Admin — full control

```
GRANT ALL ON SCHEMA MTG_SERVICING_DW.CORE TO ROLE MTG_DW_ADMIN;
GRANT ALL ON ALL TABLES IN SCHEMA MTG_SERVICING_DW.CORE TO ROLE MTG_DW_ADMIN;
GRANT ALL ON FUTURE TABLES IN SCHEMA MTG_SERVICING_DW.CORE TO ROLE MTG_DW_ADMIN;
GRANT CREATE ROLE ON ACCOUNT TO ROLE MTG_DW_ADMIN;
```

A5.4 Data Engineer — build and load, no role/user management

```
GRANT CREATE TABLE, CREATE VIEW ON SCHEMA MTG_SERVICING_DW.CORE TO ROLE
MTG_DATA_ENGINEER;
GRANT SELECT, INSERT, UPDATE, DELETE, TRUNCATE
```

```

ON ALL TABLES IN SCHEMA MTG_SERVICING_DW.CORE TO ROLE MTG_DATA_ENGINEER;
GRANT SELECT, INSERT, UPDATE, DELETE, TRUNCATE
ON FUTURE TABLES IN SCHEMA MTG_SERVICING_DW.CORE TO ROLE MTG_DATA_ENGINEER;

```

A5.5 Semantic layer — analyst-facing views

```

CREATE OR REPLACE VIEW VW_DELIQUENCY_TREND AS
SELECT d.year, d.month, b.state, k.bucket_label,
       COUNT(DISTINCT f.loan_sk) AS loan_count,
       SUM(f.unpaid_principal_bal) AS total_upb
FROM FACT_LOAN_SERVICING_EVENT f
JOIN DIM_DATE d ON f.date_sk = d.date_sk
JOIN DIM_BORROWER b ON f.borrower_sk = b.borrower_sk
JOIN DIM_DELIQUENCY_BUCKET k ON f.bucket_sk = k.bucket_sk
WHERE f.event_type = 'PAYMENT'
GROUP BY d.year, d.month, b.state, k.bucket_label;

CREATE OR REPLACE VIEW VW_ESCROW_POSITION AS
SELECT l.synthetic_loan_id, SUM(f.escrow_amt) AS net_escrow_position
FROM FACT_LOAN_SERVICING_EVENT f
JOIN DIM_LOAN l ON f.loan_sk = l.loan_sk
WHERE f.event_type = 'ESCROW'
GROUP BY l.synthetic_loan_id;

CREATE OR REPLACE VIEW VW_REMITTANCE_TIMELINESS AS
SELECT i.investor_name,
       AVG(DATEDIFF('day', f.scheduled_remit_date, f.actual_remit_date)) AS
avg_delay_days,
       COUNT(*) AS remittance_count
FROM FACT_LOAN_SERVICING_EVENT f
JOIN DIM_INVESTOR i ON f.investor_sk = i.investor_sk
WHERE f.event_type = 'REMITTANCE'
GROUP BY i.investor_name;

```

A5.6 Data Analyst — read-only via views, plus masking policy

```

GRANT SELECT ON VIEW VW_DELIQUENCY_TREND      TO ROLE MTG_DATA_ANALYST;
GRANT SELECT ON VIEW VW_ESCROW_POSITION       TO ROLE MTG_DATA_ANALYST;
GRANT SELECT ON VIEW VW_REMITTANCE_TIMELINESS TO ROLE MTG_DATA_ANALYST;
-- Note: MTG_DATA_ANALYST is NOT granted SELECT on the base tables directly –
-- self-service access is via the curated views only.

CREATE MASKING POLICY MASK_BORROWER_ID AS (val VARCHAR) RETURNS VARCHAR ->
CASE
  WHEN CURRENT_ROLE() IN ('MTG_DW_ADMIN', 'MTG_DATA_ENGINEER') THEN val
  ELSE 'MASKED-ID'
END;

ALTER TABLE DIM_BORROWER
  MODIFY COLUMN synthetic_borrower_id
  SET MASKING POLICY MASK_BORROWER_ID;

```

A5.7 Assigning roles to users

```
GRANT ROLE MTG_DW_ADMIN      TO USER GVUDAIGIRI;  
GRANT ROLE MTG_DATA_ENGINEER TO USER GVUDAIGIRI;  
GRANT ROLE MTG_DATA_ANALYST  TO USER MRCCOOPER_DEMO_ANALYST;
```

For this single-student PoC, the student's own Snowflake trial login (GVUDAIGIRI) holds both the Admin and Data Engineer roles, since both build/load and administrative tasks are performed by the same person during the academic project. A second, separate login (MRCCOOPER_DEMO_ANALYST) was created specifically to hold only the Data Analyst role, so the read-only, view-only, masked-identifier access path could be tested independently from the Admin/Engineer login rather than being asserted without verification.

References

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- Freddie Mac. Single-Family Loan-Level Dataset (schema/structure reference for synthetic-data design). <https://www.freddiemac.com>
- Fannie Mae. Single-Family Loan Performance Data (schema/structure reference for synthetic-data design). <https://www.fanniemae.com>
- Gartner & Forrester. Reports on Analytics & BI platforms and cloud data warehousing (to be cited individually as accessed).

Note: the sources above were compiled with AI assistance and have not been independently verified against their original publications. Before submission, confirm each citation's exact details — particularly the Gartner/Forrester items, which are cited generically pending specific report titles — and add any further sources drawn on during the literature review.

Glossary

- Cloud Data Warehouse: a fully managed, elastic analytical database delivered as a cloud service, with separation of storage and compute (e.g., Snowflake, BigQuery, Redshift).
- Self-Service Analytics: an approach enabling business users to access, explore, and analyse data and build reports with minimal reliance on IT/BI teams.
- Time-to-Insight: the elapsed time from a business question being raised to an answer being available.
- Concurrency: the ability of a platform to serve many simultaneous users/queries without degradation.
- UPB (Unpaid Principal Balance): the remaining principal owed on a loan at a given point in time.
- DPD (Days Past Due): the number of days a scheduled payment is overdue, used to bucket loans into delinquency stages.
- Escrow: funds collected from a borrower and held by the servicer to pay property taxes and insurance on their behalf.
- Investor Remittance: the servicer's periodic payment of collected principal/interest to the investor who owns the loan.
- Star Schema: a data-warehouse modelling pattern with a central fact table connected to descriptive dimension tables.
- Semantic Layer: a business-friendly abstraction over raw data that standardizes metrics for self-service.
- MPP (Massively Parallel Processing): an architecture that distributes queries across many nodes for faster execution.
- FinOps: practices for managing and optimizing cloud consumption cost.

Summary of how the feedback for Project Outline have been addressed

Feedback Point	How Addressed
The Supervisor's remarks on the Project Outline stated: "I have reviewed the project outline and find the topic relevant, well-scoped and aligned with the student's MBA (Business Analytics) specialization. The objectives are clear and achievable within the timeline. I am satisfied with the proposed methodology and approve the outline to proceed to the next stage." No specific revisions were requested.	As no specific changes were required, the approved scope, objectives, and mixed-methods approach were carried forward into this Mid-Semester Report unchanged. The mortgage-servicing focus, star-schema design, synthetic data generation, and role-based governance build documented in Sections 4-6 implement the methodology exactly as approved, and the Plan vs Progress table (Section 5) confirms the project remains on the timeline the Supervisor found achievable.



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes Division (WILPD)

Particulars of Student

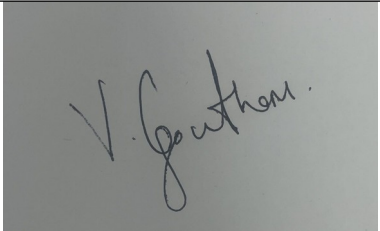
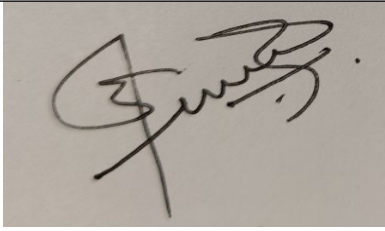
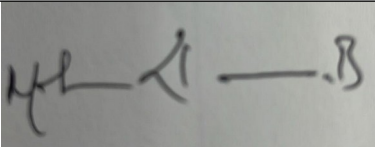
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Employing Organization and Location	NSM Services P Ltd
Programme Name	MBA in Business Analytics
Semester	Final Semester (Dissertation)
Project Title	The Role of Cloud Data Warehouses in Accelerating Self-Service Analytics in Mortgage Servicing

Particulars of the Supervisor and Additional Examiner

	Supervisor	Additional Examiner
Name	Guru Balasubramhanyan	Mohan Babu Balachandran
Qualification	BE (Electrical & Electronics), MBA. 32 years' experience (28 in the mortgage industry).	BE (Electronics & Communication Engineering). 22 years' experience in the IT industry.
Designation	SVP - Data Engineering & Analytics, Risk & Compliance	Vice President - Engineering
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Remarks of the Supervisor on Mid-Semester Project Report

I have reviewed the student's midterm progress. The Snowflake data warehouse tables have been created and populated with synthetic data. The work is progressing as planned, and I approve proceeding to the next stage of the project.

		
Signature of Student	Signature of Supervisor	Signature of Additional Examiner
Name: Gowtham Vudaigiri	Name: Guru Balasubramhanyan	Name: Mohan Babu Balachandran

Please write to WILP Project at project@wilp.bits-pilani.ac.in for any queries / clarifications.